

# BIS Sahayak (Smart India Hackathon 2026)

AI Assistant for Indian Standards & Procurement Specification Recommender (SIH26107 / SIH26108)

Live Production App: <https://sih2026-bis-assistant.vercel.app>  
Ministry: Bureau of Indian Standards & Consumer Affairs

Local Codebase: D:\SIH2026-BIS-Assistant  
Date: August 2026 (Smart India Hackathon)

## 1. Executive Summary & Problem Overview

BIS Sahayak is an intelligent, bilingual (English / Hindi), GovTech AI companion and automated specification recommender developed for the Bureau of Indian Standards (BIS) and Ministry of Consumer Affairs, Government of India.

The platform addresses two official Smart India Hackathon 2026 problem statements in a single unified architecture:

- **SIH26107:** Interactive AI assistant with dual personas (Consumer Mode vs. Technical Industry Mode), real-time clause citations, short-answer-first formatting, and Web Speech API audio synthesis.
- **SIH26108:** AI-driven Procurement & Tender Specification Recommender for automated Indian Standard discovery, match confidence scoring, and mandatory testing checklists.

## 2. Dual Personas (Consumer Mode vs. Industry Mode)

BIS Sahayak allows instant toggling between two purpose-built personas:

- **Consumer Mode:** Everyday safety guidance, counterfeit detection, 7/8-digit CM/L license checks, BIS Care App integration, and mandatory QCO consumer warnings.
- **Industry / MSME Mode:** Engineering parameters (e.g. proof stress  $\geq 500$  N/mm<sup>2</sup>, elongation  $\geq 16\%$ ), Scheme-I/IV conformity rules, Harmonized System (HS) codes, and procurement compliance.

## 3. Key Modules & Functional Capabilities

### 3.1 Conversational AI Assistant (/)

- **Short Answers First:** Provides a crisp 1-2 sentence quick answer, followed by actionable checklists and an expandable section for detailed clauses.
- **Conversational NLP & Slang:** Handles casual talk (Yo, Hi, Thanks, Who are you) naturally without dumping unsolicited technical manuals.
- **Bilingual & Audio Assist:** Instant toggle between English and Hindi with Web Speech API audio reading.

### 3.2 AI Procurement Recommender (/recommend) (SIH26108)

- **Tender Analyzer:** Paste any procurement tender or Bill of Materials (BOM). The AI extracts technical requirements and returns ranked IS standards with match confidence scores (e.g. 98%) and test checklists.

### 3.3 4-Step ISI Mark Authenticity Verifier (/verify)

- **CM/L License Validator:** Live 7/8-digit numerical license format validation with 4 visual authenticity checkpoints to identify fake products.

### 3.4 Standards Directory (/standards)

- **Searchable Catalog:** Browse 21 authentic Indian Standards across Construction, Electronics, Appliances, Safety, and Food & Water with slide-out clause inspectors.

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## 4. Indexed Indian Standards Corpus (21 Standards)

The vector database indexes 21 authentic, full-text Indian Standards:

| IS Number       | Title / Scope                               | Category              | Status & QCO Mandate      |
|-----------------|---------------------------------------------|-----------------------|---------------------------|
| IS 2347:2017    | Domestic Pressure Cookers                   | Kitchen Appliances    | Mandatory QCO (DPIIT)     |
| IS 14543:2024   | Packaged Drinking Water                     | Food & Water          | Mandatory QCO (FSSAI/BIS) |
| IS 10500:2012   | Drinking Water (Potable)                    | Food & Water          | National Standard         |
| IS 16240:2023   | RO-Based Water Treatment Systems            | Food & Water          | Mandatory QCO (MoEFCC)    |
| IS 1786:2008    | High Strength Deformed Steel Bars (Fe 500D) | Construction & Steel  | Mandatory QCO (Steel)     |
| IS 269:2015     | Ordinary Portland Cement (OPC 33/43/53)     | Construction & Steel  | Mandatory QCO (DPIIT)     |
| IS 3196 (Pt 1)  | Domestic LPG Cylinders (14.2 kg)            | Industrial & Gas      | Mandatory QCO (PESO)      |
| IS 4151:2020    | Protective Helmets for Two-Wheelers         | Consumer Safety       | Mandatory QCO (MoRTH)     |
| IS 9873 (Pt 1)  | Safety of Toys (Mechanical & Physical)      | Children & Toys       | Mandatory QCO (DPIIT)     |
| IS 15644:2006   | Safety of Electric Toys                     | Children & Toys       | Mandatory QCO (DPIIT)     |
| IS 694:2010     | PVC Insulated Cables for Wiring             | Electrical            | Mandatory QCO (DPIIT)     |
| IS 1293:2019    | Plugs and Socket-Outlets (Up to 250V)       | Electrical            | Mandatory QCO (DPIIT)     |
| IS 16102 (Pt 1) | Self-Ballasted LED Lamps (Safety)           | Electronics & IT      | Mandatory QCO (MeitY)     |
| IS 13252 (Pt 1) | Information Technology Equipment (Laptops)  | Electronics & IT      | Mandatory CRO (MeitY)     |
| IS 15885 (Pt 2) | Controlgear for LED Modules (Drivers)       | Electronics & IT      | Mandatory CRO (MeitY)     |
| IS 4984:2016    | HDPE Pipes for Potable Water                | Piping & Municipal    | Mandatory QCO (DPIIT)     |
| IS 4985:2021    | Unplasticized PVC Pipes for Potable Water   | Piping & Municipal    | Mandatory QCO (DPIIT)     |
| IS 1417:2016    | Gold and Gold Alloys, Jewellery / Artefact  | Hallmarking & Bullion | Mandatory HUID            |
| IS 2110:2019    | Silver Jewellery / Artefacts                | Hallmarking & Bullion | Voluntary Hallmark        |
| IS 16018:2021   | Lightweight Helmets for Two-Wheelers        | Consumer Safety       | Mandatory QCO (MoRTH)     |
| IS 1538:1993    | Cast Iron Fittings for Pressure Pipes       | Piping & Municipal    | Scheme-I Standard         |

## 5. Step-by-Step User Guide

- **Step 1 (AI Assistant):** Visit <https://sih2026-bis-assistant.vercel.app> -> Select Consumer/Industry mode -> Ask any product question -> Read short answer, check takeaways, or expand details.
- **Step 2 (Tender Recommender):** Go to /recommend -> Paste tender requirements or BOM -> Click Analyze -> View matched IS standards, match scores, and testing tables.
- **Step 3 (Verify ISI Mark):** Go to /verify -> Enter 7/8-digit CM/L license number -> Verify against 4 visual authenticity checkpoints.
- **Step 4 (Standards Directory):** Go to /standards -> Search by IS number or filter by sector -> Inspect full clauses and parameter limits.

## 6. Technical Architecture & Test Verification

- **Tech Stack:** Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS v4, Google Gemini API, Neon PostgreSQL pgvector.
- **Automated Testing:** 12/12 passing unit & integration tests in 226ms (Vitest).
- **Production Deployment:** Live on Vercel Serverless Production at <https://sih2026-bis-assistant.vercel.app>